

Mila Ardath

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I'm a software engineer who builds developer tools. The current one is `otel-desktop-viewer`, which shows you your OpenTelemetry data in a browser without standing up a backend. I've worked the whole stack, embedded to observability. I like building genuinely useful things, and the problems never stop being interesting.

Skills

Go, TypeScript, SQL, DuckDB, PostgreSQL, schema design, Svelte, OpenTelemetry, OTLP, gRPC, distributed tracing, observability, query language design (Lezer/CodeMirror), GitHub Actions

Experience

Open Source Maintainer, `otel-desktop-viewer` — 2023–present

github.com/CtrlSpice/otel-desktop-viewer

A local OpenTelemetry viewer: receives OTLP over HTTP and gRPC and renders traces, metrics, and logs in a browser, with no external observability platform required.

- Architected it as a custom OpenTelemetry Collector distribution: an exporter that writes telemetry, and a collector extension that owns the DuckDB store and the web UI, so storage outlives the pipeline
- Designed the analytical schema for traces, metrics, and logs, including a content-addressed attribute dictionary that dedupes attributes at ingest
- Implemented metric reduction and OTLP exponential-histogram merging in SQL, where a wrong answer still renders as a plausible chart — found four such bugs, and now verify the results against an independent implementation over randomized inputs
- Built the Svelte 5 frontend, including a search language with a Lezer grammar in CodeMirror that the Go backend compiles into SQL
- Built the release pipeline: a GitHub Actions matrix on native runners per platform (CGO rules out cross-compilation), publishing to Homebrew, GHCR, deb/rpm, and GitHub Releases across macOS, Linux, and Windows

Senior Software Engineer, Telus — 2023–2025

Internal platform organization; deployment automation for services running on Kubernetes.

- Designed and implemented the tick-based state machine at the core of a tool automating blue-green and canary deployments
- Built a golden-signal metrics comparator used as a deploy gate, so only artifacts with stable metrics promote to production
- Authored an OpenTelemetry adoption RFC and presented it to the Architecture Guild
- Technologies: Go, Kubernetes, client-go, PostgreSQL

Career break — 2016–2022

Firmware Testing and Integration Developer, Ciena — 2014–2016

- White-box firmware testing on fiber-optic network equipment, network simulation tooling to reproduce configuration errors, and thermal cycling in environmental chambers to see how switches would do in a desert
- Technologies: C, C++, TCL, Perl, Bash

Developer, Applied Research and Innovation, Algonquin College — 2012–2014

- Prototyped pro-athlete vision-training drills as a Unity game with an optometrist client, to see how they could be made accessible to more people; designed a backend-as-a-service for rapid prototyping
- Technologies: C#, Unity, Java, Android, MySQL

Education

- Recurse Center (remote) — 2025–2026
Self-directed programming retreat, admitted by application. Completed 18 weeks across two batches.
- Advanced Diploma, Engineering Technology and Computer Science, Algonquin College — 2014